

AFIF BIN MOHAMMAD TAHA

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SUMMARY

Third-year Design and Artificial Intelligence student at SUTD with a background in Infocomm and Security (NYP). I build at the intersection of hardware and software, from embedded firmware to real-time web dashboards. Experienced across the full product lifecycle including design thinking, prototyping, and deployment. Seeking internships in creative technology, design engineering, or product development.

PROJECTS

KUPPI - IoT Housekeeping Assistant

2026

SUTD Product Design Studio, Spring 2026

NFC-driven handheld device guiding hotel housekeeping staff through 6 cleaning zones, with a real-time supervisor dashboard that updates live as each zone is completed.

- Wrote all firmware in C++ on the ESP32 using FreeRTOS dual-core architecture, keeping the UI and NFC responsive while HTTP communication ran independently on the second core.
- Built full-stack supervisor dashboard: Python Flask backend, Supabase PostgreSQL database, Server-Sent Events for live room status, and a Jinja2 frontend.
- Integrated NFC reader, 3.5 inch TFT touchscreen, and LiPo battery charging circuit into a single handheld device.

Stack: C++ | Python | Flask | Arduino | FreeRTOS | Supabase | PostgreSQL | JavaScript

Wellness Buddy - Posture and Wellbeing Companion

2025

SUTD Design Thinking and Innovation, 2025

Desk companion device using OpenCV face detection to monitor posture, delivering nudges through four animated personality-driven characters built for SUTD staff working long desk hours.

- Implemented posture detection using Python and OpenCV Haar cascade classifiers for real-time face and eye tracking with distance estimation.
- Arduino-based device receives posture state over serial from the host script, triggering animated character responses and audio feedback.
- Contributed across the full project from ideation and user research to hardware assembly and software development as part of a 4-person team.

Stack: Python | OpenCV | C++ | Arduino | JavaScript

EXPERIENCE

Command School Trainer

2023 - 2024

Singapore Armed Forces

- Led and mentored groups of military trainees, regularly managing daily operations for a full company of 100 or more personnel.
- Developed communication, leadership, and structured problem-solving skills under time-pressured environments.

Desktop Engineer and Helpdesk Intern

2022

Nanyang Tech Pte Ltd

- Configured and profiled work laptops for new employees; delivered onboarding and device orientation sessions.

- Operated as walk-in helpdesk support for hardware and software issues across the organisation.

EDUCATION

BSc Design and Artificial Intelligence

2024 - 2028

Singapore University of Technology and Design (SUTD) | Minor in Computer Science and Design

Diploma in Infocomm and Security

2018 - 2022

Nanyang Polytechnic (NYP)

SKILLS

Languages: Python, C++, JavaScript, HTML, CSS, SQL

Frameworks and libraries: Flask, OpenCV, Arduino, FreeRTOS

Databases and backend: Supabase, PostgreSQL, Server-Sent Events, REST APIs

Tools: Git, GitHub, VS Code, Figma, Arduino IDE

Concepts: Embedded systems, IoT, design thinking, rapid prototyping, full-stack development